1. Computing an integral from sums

Let $a < b$. For a partition $P : a = x_0 < \cdots < x_n = b$, write $\Delta x_i = x_i - x_{i-1}$, $\|P\| = \max_i \Delta x_i$, and choose $\xi_i \in [x_{i-1}, x_i]$. Recall

$$S(f; P, \xi) = \sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

A bounded function f is Riemann integrable with integral I if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$\|P\| < \delta \quad \Longrightarrow \quad |S(f; P, \xi) - I| < \varepsilon$$

for every such partition and every choice of sample points. We use the theorem that continuous functions on closed bounded intervals are integrable.

Uniqueness. If distinct I, J both satisfied this definition, use $\varepsilon = |I - J|/3$ for each. Choose a common partition with mesh smaller than both resulting bounds and any sample points. Its sum S would satisfy

$$|I - J| \le |I - S| + |S - J| < \tfrac{2}{3}|I - J|,$$

a contradiction. Such a partition exists by taking sufficiently many equal subintervals.

Example

Compute $\int_0^1 x^2\,dx$. Since x^2 is continuous, its integral exists. Take n equal subintervals with right endpoints i/n. Their mesh is $1/n$, so their sums tend to that integral:

$$R_n = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{i}{n}\right)^2 = \frac{1}{n^3}\sum_{i=1}^{n} i^2.$$

Using $\sum_{i=1}^{n} i^2 = n(n+1)(2n+1)/6$,

$$R_n = \frac{n(n+1)(2n+1)}{6n^3} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2} \longrightarrow \frac{1}{3}.$$

Therefore

$$\boxed{\int_0^1 x^2\,dx = \frac{1}{3}.}$$

The sum formula can be checked by summing $i^3 - (i-1)^3 = 3i^2 - 3i + 1$, together with $\sum_{i=1}^{n} i = n(n+1)/2$.

The existence theorem justifies using this sequence of sums. A limit for right-endpoint sums alone would not establish integrability of an arbitrary bounded function.

2. Linearity and comparison

In Sections 2–3, let f, g be continuous on $[a, b]$, with $a < b$. All the integrals used exist by continuity.

Constant functions. Every tagged sum for $f(x) = k$ equals $\sum k\Delta x_i = k(b - a)$. Thus

$$\int_a^b k\,dx = k(b - a).$$

This also follows directly from the definition: for any $\varepsilon > 0$, choose $\delta = 1$; the error is zero.

Linearity. For constants $\alpha, \beta \in \mathbb{R}$,

$$\boxed{\int_a^b (\alpha f(x) + \beta g(x))\,dx = \alpha \int_a^b f(x)\,dx + \beta \int_a^b g(x)\,dx.}$$

Proof. Write $I_f = \int_a^b f$ and $I_g = \int_a^b g$. On the same partition with the same sample points,

$$S(\alpha f + \beta g; P, \xi) = \alpha S(f; P, \xi) + \beta S(g; P, \xi).$$

Given $\varepsilon > 0$, put $\eta = \varepsilon/(|\alpha| + |\beta| + 1)$. Choose $\delta_f, \delta_g > 0$ from the definition so the corresponding errors are less than η, and put $\delta = \min\{\delta_f, \delta_g\}$. Then $\|P\| < \delta$ implies

$$\begin{aligned} |S(\alpha f + \beta g; P, \xi) - (\alpha I_f + \beta I_g)| &\le |\alpha|\,|S(f; P, \xi) - I_f| + |\beta|\,|S(g; P, \xi) - I_g| \\ &\le (|\alpha| + |\beta|)\eta < \varepsilon. \end{aligned}$$

The definition and uniqueness prove the identity. □

Comparison. If $f(x) \le g(x)$ on $[a, b]$, then

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$$

Proof. Every width is positive, so the sums on a common tagged partition obey $S(f; P, \xi) \le S(g; P, \xi)$. Take equal partitions with right endpoints and let their mesh tend to zero. The inequality passes to the limits. Explicitly, if $I_f > I_g$, set $\eta = (I_f - I_g)/3$. For a sufficiently fine common partition, $S(f) > I_f - \eta > I_g + \eta > S(g)$, a contradiction. □

Example

By Section 1 and linearity,

$$\int_0^1 (4 + 3x^2)\,dx = 4 + 3 \cdot \frac{1}{3} = 5.$$

3. Bounds and additivity

Bounds. Comparison with constant functions gives

$$m \le f(x) \le M \quad (a \le x \le b) \quad \Longrightarrow \quad m(b-a) \le \int_a^b f(x)\,dx \le M(b-a).$$

Also, $-|f| \le f \le |f|$, so comparison and linearity give

$$\boxed{\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx.}$$

In particular, if $|f(x)| \le M$, then $\left|\int_a^b f(x)\,dx\right| \le M(b-a)$.

Additivity. If $a < c < b$, then

$$\boxed{\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.}$$

Proof. Divide each of $[a,c]$ and $[c,b]$ into n equal pieces and choose any sample points. Their union is a partition of $[a,b]$ with mesh $\max\{(c-a)/n, (b-c)/n\}$, which tends to zero. The sum on the union is exactly the sum of the two smaller sums. All three integrals exist by continuity, so taking limits gives the identity. □

Orientation. Define

$$\int_a^a f(x)\,dx = 0, \qquad \int_b^a f(x)\,dx = -\int_a^b f(x)\,dx.$$

With these conventions, additivity holds for any ordering of the three endpoints, provided f is continuous on an interval containing them. In particular, for any u, v in that interval,

$$\boxed{\left|\int_u^v f(t)\,dt\right| \le \int_{\min(u,v)}^{\max(u,v)} |f(t)|\,dt \le M|v-u|}$$

whenever $|f| \le M$ between u and v. For $v < u$, reverse the bounds before applying comparison.

Example

On $[0,1]$, $1 \le e^{x^2} \le e$. Consequently,

$$1 \le \int_0^1 e^{x^2}\,dx \le e.$$

This estimate does not require a formula for an antiderivative.

4. The Fundamental Theorem of Calculus, part I

Theorem. Let $a < b$ and let f be continuous on $[a, b]$. Set

$$A(x) = \int_a^x f(t)\,dt \qquad (a \le x \le b).$$

Then A is continuous on $[a, b]$, differentiable on (a, b), and $\boxed{A'(x) = f(x)}$ for $a < x < b$.

Proof. By the Extreme Value Theorem, $|f(t)| \le M$ on $[a, b]$ for some $M \ge 0$. Additivity and the oriented bound give

$$|A(y) - A(x)| = \left|\int_x^y f(t)\,dt\right| \le M|y - x|.$$

This proves continuity, including at the endpoints: given $\varepsilon > 0$, use $|y - x| < \varepsilon/(M + 1)$.

Fix $x \in (a, b)$. For $h \ne 0$ with $x + h \in [a, b]$, additivity, linearity and the constant integral give

$$\frac{A(x + h) - A(x)}{h} - f(x) = \frac{1}{h}\int_x^{x+h} (f(t) - f(x))\,dt.$$

Given $\varepsilon > 0$, continuity at x gives $\delta_0 > 0$ such that $|f(t) - f(x)| < \varepsilon/2$ when $|t - x| < \delta_0$ and $t \in [a, b]$. Set $\delta = \min\{\delta_0, x - a, b - x\} > 0$. If $0 < |h| < \delta$, every t between x and $x + h$ satisfies this bound. Hence

$$\begin{aligned}\left|\frac{A(x + h) - A(x)}{h} - f(x)\right| &\le \frac{1}{|h|}\int_{\min(x,x+h)}^{\max(x,x+h)} |f(t) - f(x)|\,dt \\ &\le \frac{1}{|h|}\frac{\varepsilon}{2}|h| = \frac{\varepsilon}{2} < \varepsilon.\end{aligned}$$

Thus $A'(x) = f(x)$. The argument includes both signs of h. □

The theorem constructs an antiderivative of a continuous function. A definite integral with a variable endpoint can therefore define a function even when no explicit antiderivative formula is available.

Example

Let $A(x) = \int_0^x e^{t^2}\,dt$ for $x \in \mathbb{R}$. Applying the same argument on a finite interval containing 0 and x gives

$$A'(x) = e^{x^2}, \qquad \frac{d}{dx}\left(\int_0^{x^2} e^{t^2}\,dt\right) = \frac{d}{dx}A(x^2) = 2xe^{x^4}.$$

The integration variable is t; x enters through the endpoint. The formula with upper endpoint x^2 follows from the chain rule.

5. The Fundamental Theorem of Calculus, part II

Theorem. Let $a < b$ and let f be continuous on $[a, b]$. Suppose F is continuous on $[a, b]$, differentiable on (a, b), and $F'(x) = f(x)$ for $a < x < b$. Then

$$\boxed{\int_a^b f(x)\,dx = F(b) - F(a).}$$

We write $[F(x)]_a^b$ for the endpoint difference $F(b) - F(a)$.

Proof. Part I gives a function $A(x) = \int_a^x f(t)\,dt$, continuous on $[a, b]$, with $A' = f$ on (a, b). Thus $H = F - A$ is continuous on $[a, b]$, differentiable inside, and $H' = 0$. The Mean Value Theorem on $[a, b]$ gives $H(b) - H(a) = 0$. Therefore

$$F(b) - A(b) = F(a) - A(a).$$

Since $A(a) = 0$, rearranging proves the formula. □

The definite integral retains its definition by Riemann sums. The theorem now permits its evaluation from any antiderivative that meets the hypotheses. Adding a constant to F does not change $F(b) - F(a)$.

Example

For $F(x) = x^3/3$, we have $F' = x^2$, so

$$\int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}.$$

This agrees with the sum calculation in Section 1. Since $(\ln x)' = 1/x$ on $(0, \infty)$,

$$\int_1^e \frac{1}{x}\,dx = [\ln x]_1^e = 1.$$

Hypotheses. There is no Riemann integral of $1/x$ on $[-1, 1]$: the expression is undefined at 0 and is unbounded near it. Assigning a value at zero would not make it bounded. The endpoint values of $\ln|x|$ do not justify applying the theorem across zero. They belong to antiderivatives on two separate intervals, neither containing all of $[-1, 1]$.